Part 2

Part 2a

We are given the equation for $\Psi(x,t)$:

$$\Psi(x,t) = A\frac{1}{\sqrt{1+\frac{2i\hbar at}{m}}}\exp\left(\frac{-ax^2+ik_0x-i\frac{\hbar k_0^2}{2m}t}{1+\frac{2i\hbar at}{m}}\right)$$

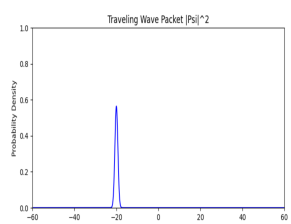
We are told to verify that $\Psi(x,0) = Ae^{-ax^2}e^{ik_0x}$, which can be verified simply by plugging $t=0$ into the original equation for $\Psi(x,t)$.

$$\Psi(x,0) = A\frac{1}{\sqrt{1+0}}\exp\left(\frac{-ax^2+ik_0x-0}{1+0}\right) = A(1)\exp\left(-ax^2+ik_0x\right) = Ae^{-ax^2+ik_0x} = Ae^{-ax^2}e^{ik_0x}$$

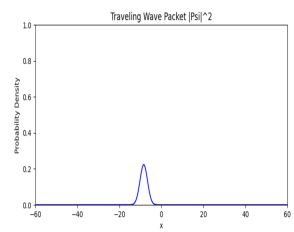
Part 2b and 2c

For the computational setup of the animation, I chose to have $N = 2001$, $x_{min} = -60$ (meters), $x_{max} = 60$ (meters),$x_0 = -20$ (meters), $m = 10^{-34}$ (kg), and the amount of time that advanced per frame ('dt' in a certain sense) was set to be 0.05 seconds.

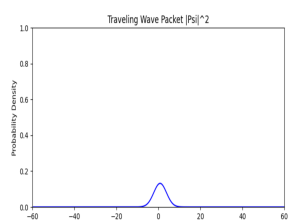
Here are images at $t = 0$, $t \approx 2$, $t \approx 4$, and $t \approx 6$ seconds



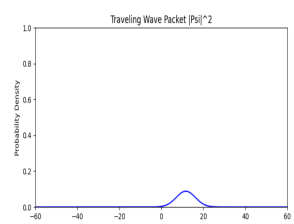
(a) $t = 0$ seconds



(b) $t \approx 2$ seconds



(c) $t \approx 4$ seconds



(d) $t \approx 6$ seconds

We can clearly see that the probability function propagages to the right and disperses as well, as time increases. The group velocity, or velocity of the expected x (in this case, the peak of the probability function) appears to be approximately 5 meters per second given these initial parameters.